Agentic Bug Triage & Routing

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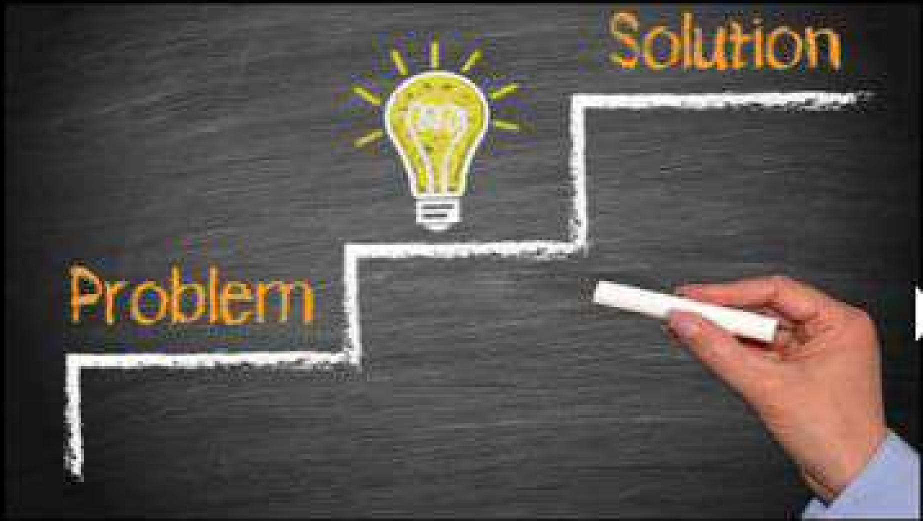


Project Details

Theme/Title: Agentic Bug Triage & Routing

- Objective/Abstract:

Modern engineering organizations — including HPE — use multiple, disconnected bug-tracking systems such as on-prem JIRA, cloud JIRA, Bugzilla, and other custom MCP-backed repositories. This fragmentation creates several systemic challenges:

Scattered data: Engineers must search across several systems to understand the true state of a bug. There is no single pane of glass.

Manual triage & routing: Incoming bugs often end up in the wrong repository, forcing engineers to manually reclassify and re-file them. This wastes time and increases error rates.

Lost lineage & context: When a bug is copied or reclassified into another system, the relationship between the original and the new ticket is often not maintained — leading to inconsistent tracking across products.

Operational overhead: Engineers spend significant effort switching contexts across multiple tools, which slows down remediation and degrades customer experience.

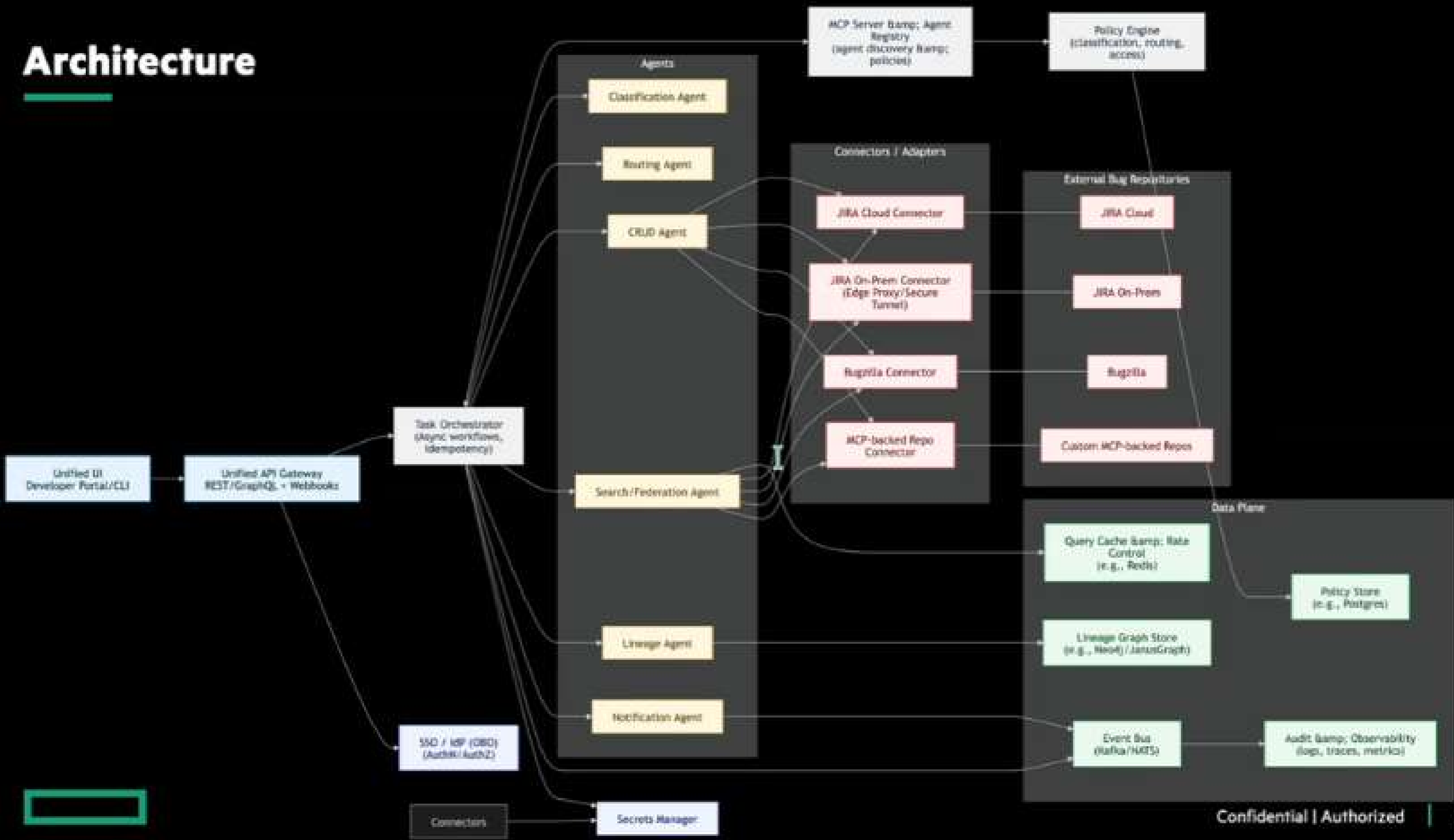
- Proposed Solution:

The solution is to build an agentic, multi-repository bug management system that acts as a single unified interface for all bug-related workflows — create, read, update, delete — irrespective of where the bug "actually lives." The solution uses a combination of multi-agent automation, policy-based routing, and lineage tracking: A Single Interface One UI/API to perform all bug operations across all repositories (JIRA on-prem, JIRA cloud, Bugzilla, MCP-backed repos).

This eliminates context switching and unifies visibility. Automated Classification & Routing Agents analyze new bug submissions and determine: The correct lead repository, Whether the bug should be reclassified to another system, Whether multiple product repositories must be updated simultaneously. Real-Time Multi-Repo Search Search across all bug repositories in real time — a foundational step before CRUD automation.

Lineage Preservation When a bug is copied or routed elsewhere, the system automatically updates the original record with lineage links so all stakeholders have consistent visibility. Multi-Agent Architecture Following HPE's broader agentic AI approach, this solution operates as part of a multi-agent ecosystem, integrating with MCP servers and the agent registry for search, CRUD, and orchestration.

Architecture



Technology

1. AI/ML
2. Python, PyTorch, Jupyter Lab, Google Collab
3. Data Storage fundamentals
4. Virtualization
5. Linux

Project Details and References

- Development / Runtime environments
  - Programming language: Python, Jupyter Lab, C++
  - Public Cloud: Lab environment

- Expected outcomes
  - Project report
  - Demonstrate a functional solution that can for Agentic Bug Triage.

References:

- https://pypi.org/
- https://www.kaggle.com/
- https://wasabi.com/learn/
- Https://deeplearning.ai